

SECSR2043 OPERATING SYSTEMS
[20 Marks]

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Marks

Instruction: Please answer all of the following questions. Whenever the □ symbol appears, please raise your hand to call your instructor, he/she will verify your results by putting his / her initial next to the symbol.

1. Type the following commands using a text editor and save it as a *yourname.sh* (Example: ahmad.sh).

```
echo "Hello world" > helloworld.jar
mkdir cars; mkdir dates; mkdir fruits drinks
cd cars; echo "Honda Accord" > accord.c
cp accord.c civic.c; echo proton > proton.c; cd
../dates;
date > dateoftheday
cat dateoftheday > appointment
cd ../fruits; echo apple > apple.txt; cat apple.txt >
orange.txt
cd drinks; cp ../cars/*.c .; cp ../fruits/*.c .;
cp ../*.jar .
```

- a) Execute the script and draw a tree structure that contains created directories and files. The parent node of the directory begin with **\$HOME** directory.

[4 marks]



Print screen the script that you type;

```

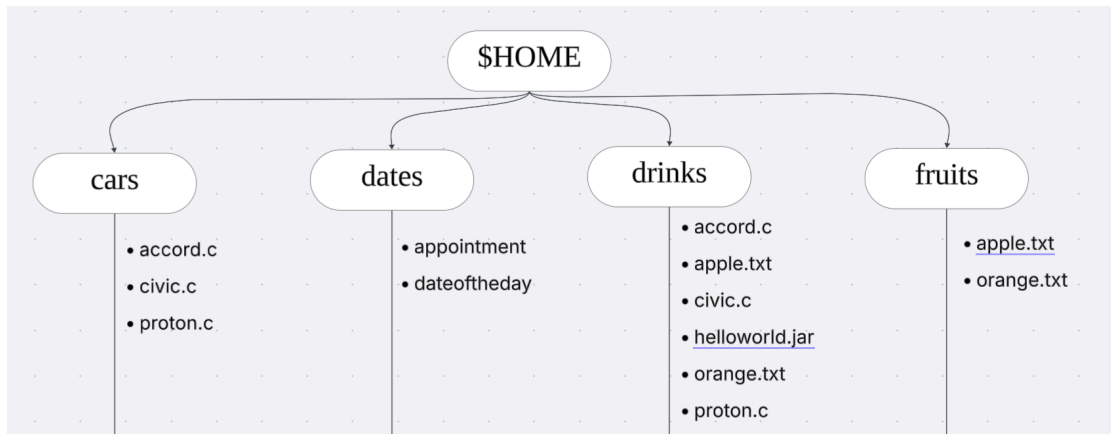
GNU nano 7.2                                nuraisyah.sh
echo "Hello world" > helloworld.jar
mkdir cars; mkdir dates; mkdir fruits drinks
cd cars; echo "Honda Accord" > accord.c
cp accord.c civic.c; echo proton > proton.c; cd ../dates;
date > dateoftheday
cat dateoftheday > appointment
cd ../fruits; echo apple > apple.txt; cat apple.txt > orange.txt
cd drinks; cp ../cars/*.c .; cp ../fruits/*.c .;
cp ../*.jar .
  
```

Then draw the tree

```

nuraisyah@secr2043:~$ tree
.
|-- cars
|   |-- accord.c
|   |-- civic.c
|   `-- proton.c
|-- dates
|   |-- appointment
|   `-- dateoftheday
|-- drinks
|   |-- accord.c
|   |-- apple.txt
|   |-- civic.c
|   |-- helloworld.jar
|   |-- orange.txt
|   `-- proton.c
|-- fruits
|   |-- apple.txt
|   `-- orange.txt
|-- helloworld.jar
`-- nuraisyah.sh

5 directories, 15 files
  
```



- b) Write an interactive bash script that will read a type of file extension, display all those files, and count the number of files. To validate your script, display c program files, and enter “c” as the input to the bash script. [4 marks]

```

GNU nano 7.2 countfiles.sh
#!/bin/bash

read -p "Enter the file extension (without the dot): " ext

echo "Files with .${ext} extension: "
find . -type f -name "*.${ext}"

count=$(find . -type f -name "*.${ext}" | wc -l)
echo "Total files with .${ext} extension: $count"

```

```

nuraaisyah@secr2043:~$ chmod u+x countfiles.sh
nuraaisyah@secr2043:~$ ./countfiles.sh
Enter the file extension (without the dot): c
Files with .c extension:
./drinks/civic.c
./drinks/accord.c
./drinks/proton.c
./cars/civic.c
./cars/accord.c
./cars/proton.c
Total files with .c extension: 6

```

2. The following Figure 1 illustrates a tree structure of some directories and files.

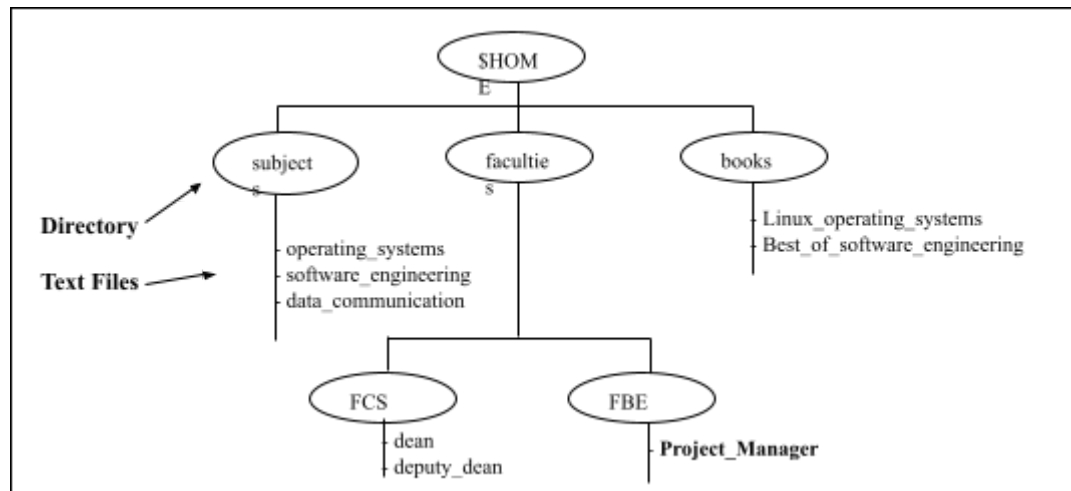


Figure 1

- a) Write a bash script (called `myname2a.sh`) that will produce directories and files as in Figure 1. Each text files contain its filename without the underscore character. For example: text file `Project_Manager` contains `Project Manager`). [4 marks]

```

GNU nano 7.2                                nuraisyah2a.sh
#!/bin/bash

mkdir -p $HOME/subjects $HOME/faculties/FCS $HOME/faculties/FBE $HOME/books

echo "Operating Systems" > $HOME/subjects/operating_systems
echo "Software Engineering" > $HOME/subjects/software_engineering
echo "Data Communication" > $HOME/subjects/data_communication

echo "Dean" > $HOME/faculties/FCS/dean
echo "Deputy Dean" > $HOME/faculties/FCS/deputy_dean

echo "Project Manager" > $HOME/faculties/FBE/Project_Manager

echo "Linux Operating Systems" > $HOME/books/Linux_operating_systems
echo "Best of software engineering" > $HOME/books/Best_of_software_engineering
  
```

```
nuraisyah@secr2043:~$ chmod u+x nuraisyah2a.sh
nuraisyah@secr2043:~$ ./nuraisyah2a.sh
nuraisyah@secr2043:~$ tree $HOME
/home/nuraisyah
|-- books
|   |-- Best_of_software_engineering
|   `-- Linux_operating_systems
|-- cars
|   |-- accord.c
|   |-- civic.c
|   `-- proton.c
|-- countfiles.sh
|-- dates
|   |-- appointment
|   `-- dateoftheday
|-- drinks
|   |-- accord.c
|   |-- apple.txt
|   |-- civic.c
|   |-- helloworld.jar
|   |-- orange.txt
|   `-- proton.c
|-- faculties
|   |-- FBE
|   |   |-- Project_Manager
|   `-- FCS
|       |-- dean
|       `-- deputy_dean
|-- fruits
|   |-- apple.txt
|   `-- orange.txt
|-- helloworld.jar
|-- nuraisyah.sh
|-- nuraisyah2a.sh
`-- subjects
    |-- data_communication
    |-- operating_systems
    `-- software_engineering

10 directories, 25 files
nuraisyah@secr2043:~$
```

- b) Complete the following table by writing the access control of directories or files that were produced. Given is the access control for directory called book. [2

marks]

Directory/File	Access Control
books	drwxrwxr-x
subjects	drwxr-xr-x
Best_of_software_engineering	-rw-r--r--
FCS	drwxr-xr-x
project_manager	-rw-r--r--

```
drwxr-xr-x 2 nuraisyah nuraisyah 4096 Jun 16 04:31 subjects
```

```
-rw-r--r-- 1 nuraisyah nuraisyah 29 Jun 16 08:28 Best_of_software_engineering
```

```
drwxr-xr-x 2 nuraisyah nuraisyah 4096 Jun 16 04:31 FCS
```

```
-rw-r--r-- 1 nuraisyah nuraisyah 16 Jun 16 08:28 Project_Manager
```

- c) Write another bash script (called `myname2c.sh`) that will change the access control of the directories and files based on the following information:

[4

marks]

Directory/File	Users								
	Owner			Group			Public		
subjects					X	X		X	X
Best_of_software_engineering		X		X		X	X	X	X
FCS			X	X	X	X			
project_manager	X	X	X	X			X	X	

```

GNU nano 7.2                                nuraisyah2c.sh
#!/bin/bash
chmod 551 $HOME/subjects

chmod 664 $HOME/books/Best_of_software_engineering

chmod 771 $HOME/faculties/FCS

chmod 115 $HOME/faculties/FBE/Project_Manager

```

- d) Complete the following table by writing the access control for each directory or file after executing the bash script in question 2(c)).

[2

marks]

Directory/File	Access Control
subjects	<code>dr-xr-x--x</code>
Best_of_software_engineering	<code>-rw-rw-r--</code>
FCS	<code>drwxrwx--x</code>
project_manager	<code>---x--xr-x</code>

```

dr-xr-x--x 2 nuraisyah nuraisyah 4096 Jun 16 04:31 subjects
-rw-rw-r-- 1 nuraisyah nuraisyah 29 Jun 16 08:28 Best_of_software_engineering
drwxrwx--x 2 nuraisyah nuraisyah 4096 Jun 16 04:31 FCS
---x--xr-x 1 nuraisyah nuraisyah 16 Jun 16 08:28 Project_Manager

```

End of Lab 3

**** All the Best for Final Exam ****